Vocabulary position and trademark lifecycles:
An event-dated corpus and a lead/lag text measure for the commercial economy

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Abstract

Text-based novelty measures built on patents observe invention by the small minority of firms that patent. Trademark filings time-stamp commercialization by everyone else, in plain commercial language.

All 13.99 million USPTO trademark case files are re-parsed into event-dated prosecution histories, and each filing scored on one idea: how surprising its language is. *Atypicality* asks how surprising the description would have been to a reader who knew only what the industry had filed before it. *Forward lean*, written ΔKL, asks the same question from the other side — how surprising it looks to a reader who also knows what the industry filed afterwards. A filing that was surprising then and ordinary later was early, and was copied.

Conditioning on marks the office granted, surprising language marks firms that reach public markets: within class and filing year, and holding description length fixed, debut owners in the top atypicality quintile reach the SEC-listed universe about a quarter more often than those in the bottom, on a 0.19% base rate ($t = 3.8$). Among surprising filings, having been early is what distinguishes the failures. Forward-leaning marks fail their first Section 8 use-proof gate 2.3 percentage points more often per within-class-cohort quintile, on 3.10 million registrations against a 49% base failure rate, with class×cohort fixed effects, owner-clustered errors, and drafting and filer controls ($t = 7.8$); the penalty emerges with the 2008–2010 cohorts and reproduces out of sample in 2019. At equal atypicality, forward-leaning debuts reach listing *less* often ($t = -5.0$), and less often still among firms that raise venture financing. Being surprising pays; having been copied does not. The measure exists for the millions of firms that never patent and is essentially uncorrelated with patenting within firm.

Two cautions bound how far this travels. Read without conditioning on grant, the same data support weaker and sometimes opposite conclusions, because registration is itself selected on language. And scored on individual terms rather than topic distributions, forward-leaning filings appear to belong to better firms — an association that dissolves or reverses under distribution scoring, because term scoring picks up lexical specificity rather than position. Text measures of this kind can manufacture firm-performance correlations out of drafting style. Code, panels, and the corpus build are public at `https://github.com/ericsilver/novelty`.

1 Introduction

Text-based measurement has reshaped how economists quantify invention. Kelly et al. (2021) score every US patent since 1840 by its textual distance from prior and similarity to subsequent patents;

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Kalyani (2024) measures patent creativity as the share of technical terminology not seen before and ties it to firm productivity. Both programs inherit the patent record's coverage: technical invention by the small minority of firms that patent. Most firms never patent, service-sector and business-model innovation rarely patents, and patenting propensity varies so much across industries that cross-industry comparisons are fraught (Dinlersoz et al., 2018; Graham et al., 2013). Trademark filings offer the complementary window. Nearly every commercially serious product or service introduction in the United States is accompanied by a trademark filing, the filing's goods/services description states in plain commercial language what the firm intends to sell, and the corpus extends across every sector of the economy at the rate of hundreds of thousands of granted filings per year.

The trademark–innovation literature is three decades old and has moved well past counting. Flikkema et al. (2025) survey its first ten years of validation work and its open questions; Willeke et al. (2026) inventory the data sources and the preprocessing burden that has kept them underused. Micro-level links to firm dynamics are established (Dinlersoz et al., 2018, 2023), and the goods/services description is already read as text rather than treated as a label: Semadeni (2006) scores consulting service-mark descriptions into a competitive positioning space, Semadeni & Anderson (2010) tracks whether one firm's described offering is imitated by rivals, and Flikkema et al. (2019) codes descriptions against the innovation each mark accompanies. Prosecution records have been exploited too: Fink et al. (2024) use contested filing sequences to study trademark races, and their outcome — continued use five years after registration — is the margin this paper's central result also runs on. What Flikkema et al. (2025) put as the field's next step is the move from counts and innovation intensity to the *type*, *quality*, and *trend* content of what marks cover: the trademark analogue of what Kelly et al. (2021) and Kalyani (2024) did for patent text. That analogue does not yet exist at corpus scale, and the two strands above — description text and prosecution events — have run on separate data.

This paper builds both on one corpus. The first contribution is that corpus: all 13.99 million USPTO trademark case files re-parsed into event-dated prosecution histories — every office action, response, maintenance filing, and cancellation with its date, joined to counsel of record, filing basis, and owner domicile. Where the patent record dates invention once (at grant), the trademark prosecution record dates a commercial claim's entire life, including the Section 8 use-proof gates at which marks attached to products that never materialized are culled. The second contribution is a filing-level text measure defined on that corpus. The third is a measurement hazard, documented at corpus scale, that applies to text-based novelty measures generally.

The procedure measures one thing: how surprising a filing's language is. Group every trademark filing by its industry class — one of the 45 international goods and services categories — and by its year, then ask how surprising its wording would have been to a reader who knew only what that industry had filed over the previous five years. That is *atypicality.* Ask the same question from the other side — how surprising the wording looks to a reader who has *also* seen what the industry filed over the following five years — and the answer usually differs. The gap between the two is *forward lean*, written ΔKL. Each "how surprising" is a Kullback–Leibler divergence between the filing's mix of topics and its industry's, large when the filing puts weight where its industry's filings put little.

What the gap captures is worth stating plainly. Language that was surprising when filed and unremarkable five years later was surprising only because it came first: the industry moved toward it, which is to say the filer was early and was copied. Language that read as ordinary when filed and unusual later was behind its industry rather than ahead of it. So atypicality asks whether a filing said something surprising, and forward lean asks whether the surprise was merely a head start. Both are computable for every filing in the corpus, including the millions from firms that never patent.

Trademark law then supplies an unusually direct test, because a registered mark must be re-proved in use five to six years on or be cancelled — a dated, adversarial checkpoint on whether the product the mark named ever reached the market.

The construct adapts the prospective/retrospective Kullback–Leibler framework that Murdock et al. (2017) developed on Darwin's reading notebooks and Barron et al. (2018) extended to French Revolution parliamentary debates. Two departures matter. There is no single authoring agent here: each filing is scored against the stream of *other* firms' filings in its class, so what is measured is a filing's position in its industry's vocabulary, not an author's cognitive novelty. And legal drafting rewards caution about convention, so what looks like surprise in boilerplate may be style, length, or counsel effects — which is why the design leans on fixed effects, drafting covariates, and owner clustering rather than raw gradients.

The nearest measurement precedent is von Graevenitz et al. (2022), who flag goods/services tokens that are both new and fast-spreading and use their geography to show that distance still retards the diffusion of innovations. Their unit is the token and their novelty flag is binary, assigned once a token's diffusion path is known; ΔKL is a continuous score for the *filing*, so every application carries a position, not only those containing a flagged word, and that position can be crossed with the mark's own prosecution outcomes.

The central results are stated on marks the office actually granted.[1] That conditioning is not incidental. Registration is itself selected on language, so a correlation computed over all filings mixes what the examination process does to a description with what the market does to the product. Appendix C reports the unconditional version and shows where it diverges.

Among granted marks, surprising language marks firms that reach public markets: atypical debut filings belong disproportionately to owners who later appear in the SEC-listed universe. Conditional on being surprising, having been early is what separates the failures. Forward-leaning marks fail the use-proof gates more often, and at equal atypicality their owners list less often, not more. Being copied, in other words, is the part of surprise that does not pay. The risk side survives a design built to kill it — fixed effects, drafting and filer controls, owner clustering, and estimation inside the counsel-represented US-domiciled stratum.

Section 2 walks through the life of a trademark and the sample the funnel defines. Section 3 defines the measure. Section 4 reports the two-axis results, the patent contrast, and a staged table (Section 4.5) that re-estimates every outcome on one sample, from registration through venture financing to public listing. Section 5 applies the corpus to cross-industry vocabulary diffusion. Section 6 states limits and the research agenda. Computation details, scoring robustness, and the measurement-hazard forensics are in the appendices.

2 Registration measures follow-through; the Section 8 gate measures survival

A US trademark passes through a sequence of dated, high-stakes steps, and each step leaves a record. An application is filed with a goods/services description; an examiner reviews it, often issuing office actions the applicant must answer; if allowed and (for intent-to-use filings) a statement of use is filed, the mark registers. Registration starts a maintenance clock: in years five to six the owner

[1] *Atypicality* is used here in a purely textual sense — the divergence between a filing's goods/services description and the vocabulary of its Nice class in that year — and not in the sense of trademark law's distinctiveness doctrine, which governs a mark's registrability along the generic-to-fanciful spectrum and under §2(f). The two constructs attach to different objects: legal distinctiveness is a property of the mark, while the quantity measured here is a property of the description of the goods and services the mark covers.

must file a Section 8 declaration proving continued use in commerce, with a six-month grace period; around year ten a combined Section 8 declaration and Section 9 renewal is due, and every ten years thereafter. A mark that misses a use-proof gate is cancelled. Madrid Protocol registrations (§66(a) basis) file Section 71 declarations on the same schedule and are flagged separately throughout. Graham et al. (2013) describe the administrative data; the event-dated corpus built here adds the full prosecution history of each case, so a cancellation is assigned to a specific gate by its date rather than inferred from a status snapshot.

Table 1: The trademark funnel in this corpus. Registration counts are unique serials; gate outcomes are event-dated (a Section 8 cancellation at registration age 4.0–8.5 years). Sources: the TRTYRAP re-parse and the scored filing panels described in Section 3.

Stage	n	Share
Debut filings scored, 1995–2018 (owners' first filings)	3,589,421	
reached registration	2,066,855	57.6%
of failures: never filed statement of use		42.6%
of failures: stopped responding in prosecution		45.3%
of failures: removed adversarially		4.3%
Registrations 2002–2018, scored, unique serials	3,104,951	
failed the first Section 8 gate (event-dated)		49.0%
Passed first gate, cohorts 2002–2013, terminal status known	1,130,082	
failed at the year-ten renewal		39.1%

The funnel (Table 1) fixes the paper's reading of each step. Registration is a launch indicator rather than an examiner verdict: among applications that fail to register, roughly nine in ten die by the applicant's own inaction — never filing the statement of use that registration requires, or going silent during prosecution — and only 4.3% are removed by opposition, appeal, or other adversarial proceedings. Completing registration therefore mostly measures whether the filer followed through to actual commercial use. The Section 8 gate is the marketplace speaking five years later: half of registered marks do not survive it, and failing it means the product or service the mark named is no longer (or never was) in commerce.

All lifecycle results below condition on this funnel once. The registration step is analyzed as its own outcome; the gate results condition on registration, so they compare marks that launched; and the unconditional composite — which mechanically stamps the registration step's shape onto any downstream outcome — is confined to Appendix C. Outcomes are read two ways throughout: from the USPTO status-code dictionary,[2] ground-truthed against marks with publicly known fates, and, for the primary results, from the event-dated prosecution history, which attributes each cancellation to a specific gate and observes counsel and basis directly.

3 Two axes: how atypical a filing's language is, and which way it leans

Kullback–Leibler divergence is a distance-like quantity between two distributions: it is large when a filing puts weight where its industry's filings put little, and near zero when the two look alike. It therefore measures surprise, and the reference distribution decides whose surprise. Measured against what the industry filed *before*, it says how surprising the filing was to a reader at the time.

[2] 700 registered; 701–705 Section 8 accepted; 710 cancelled for Section 8 failure; 800 renewed; 900 expired.

Measured against what the industry filed *after*, it says how surprising the same wording looks to a reader who knows what came next. Positive ΔKL — surprising then, ordinary later — marks a filing the field moved toward.

Formally, for a time-ordered sequence of documents, with each document at year t producing a distribution P_t, the framework defines two quantities against a window W of past and future documents (Barron et al., 2018; Murdock et al., 2017):

- *Prospective surprise*: $\mathrm{KL}(P_t \,\|\, Q_{\text{past}})$, where Q_{past} aggregates documents in years $t - W$ to $t - 1$;
- *Retrospective surprise*: $\mathrm{KL}(P_t \,\|\, Q_{\text{future}})$, where Q_{future} aggregates years $t + 1$ to $t + W$.

The signed difference is

$$\Delta\mathrm{KL}_t \;=\; \mathrm{KL}(P_t \,\|\, Q_{\text{past}}) - \mathrm{KL}(P_t \,\|\, Q_{\text{future}}).$$

Negative ΔKL marks language that looked normal at filing and played-out in hindsight. The KL *level* against the past is *atypicality*: how surprising the filing was when written. The signed difference is *forward lean*: how much of that surprise was a head start rather than a difference the industry never adopted. The results below use both, because they turn out to carry different information.

Scoring runs on topic distributions rather than raw tokens, for one reason. Goods/services descriptions typically run 10–30 words, and on documents that short, token-level KL inflates mechanically: mean |ΔKL| varies 2.5-fold across document-length bands. Scoring each filing's topic distribution instead gives every filing a dense representation regardless of length, and cuts that length gradient to about 1.4-fold. Fitting details, the resolution sweeps at $T = 200$ and $T = 500$, the term-level variant, the 1995 burn-in cut, per-filing reliability, and window robustness are all in Appendix A; the headline results hold across an order of magnitude in topic resolution and across representations.

The choice of distributions over terms is not aesthetic. Term-level scoring additionally captures lexical specificity — rare words used by serious firms describing specific products — and that channel manufactures signed firm-performance correlations that distribution scoring exposes as artifacts. Appendix B documents the hazard in full; it is the reason every headline number in this paper is reported under distribution scoring with term scoring as the cross-check, and it is a caution portable to any text-based novelty measure estimated at the term level.

Figure 1 grounds the construct in recognizable examples. The two axes are prospective and retrospective surprise, with ΔKL the diagonal residual: below the diagonal, language that looked unusual at filing and became standard (OpenAI, Instagram, White Claw); above it, language the field moved away from (Smartwater 1997, Celsius 2004).

4 Among granted marks, surprise pays and being early does not

4.1 Why these results condition on grant

Registration is not a neutral filter on language. Across 3.59 million debut filings (57.6% base rate), completion is an inverse-U in ΔKL: the middle quintiles complete best (∼58.6–58.9%) and both tails complete less (54.5% at the bottom, 56.8% at the top; Figure 2A). The shape is scoring-robust and near-universal (positive middle-versus-tails depth in 37 of 44 classes under term scoring, statistically distinguishable in 33). Applicants filing category-typical language follow through to commercial use most reliably; both kinds of vocabulary bet fail to launch more often.

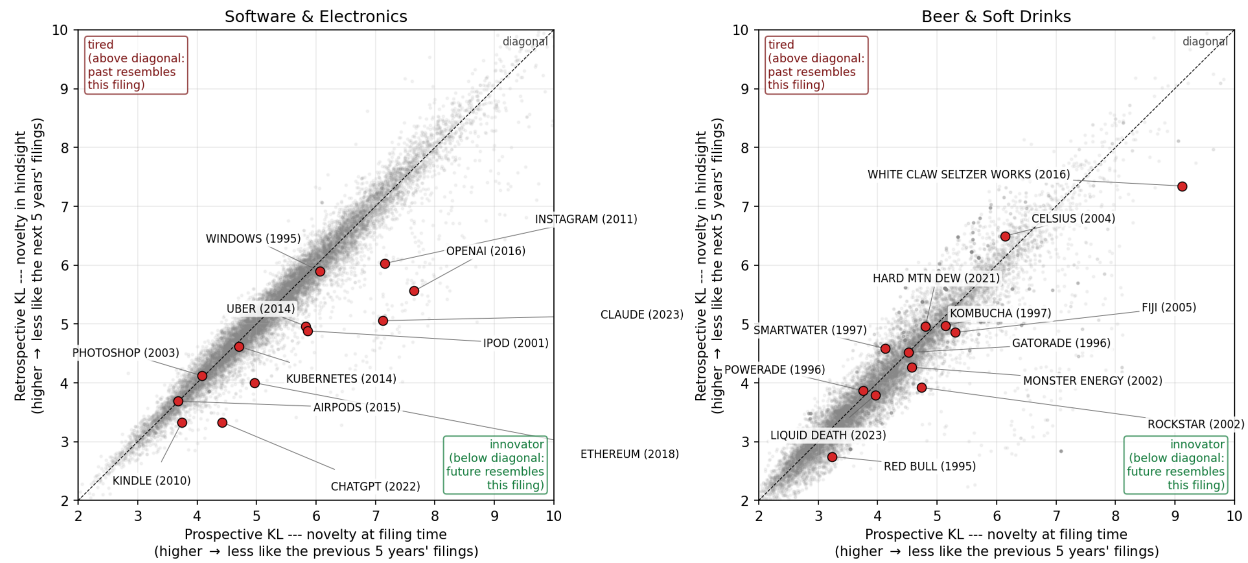

Figure 1: Prospective KL versus retrospective KL with ΔKL as the above-diagonal/below-diagonal residual. Labeled examples are recognizable post-burn-in brand introductions, positioned by their goods/services text. *Left:* software/electronics. OpenAI (+2.09), Claude (+2.06), Instagram, ChatGPT, iPod, Ethereum, and Uber sit on the innovator side; Photoshop (2003) and AirPods sit essentially on the diagonal. *Right:* a beer/soft-drinks diagnostic class. White Claw (+1.78), Rockstar, and Red Bull lean innovator; Smartwater (1997) and Celsius (2004) lean tired.

That is a result about the examination funnel, not about the market, and it is why everything below is estimated on marks the office granted. Pooling granted and abandoned filings mixes the two selections, and the resulting gradients are attenuated and in places reversed relative to the conditional ones. Appendix C reports that comparison in full; it belongs with the measurement cautions rather than with the findings.

4.2 Forward-leaning marks fail the use-proof gate

Marks written in language their industry subsequently moved toward fail their first Section 8 use-proof gate more often than marks written in language it was moving away from. Within class×cohort cells, with owner-clustered errors and drafting and filer controls, the failure rate rises +2.3pp per ΔKL quintile against a 49% base rate ($t = 7.8$). This is the paper's central result, and the rest of this section is the design that produces it.

Raw pooled quintiles convey the direction: among 4.12 million registration class-records 2002–2018, dated first-gate cancellation rises from 43.7% at the bottom topic-ΔKL quintile to 47.7% at the top, positive in 31 of 44 classes with sufficient volume (Figure 3). But a raw pooled contrast mixes the within-industry gradient with class and cohort composition, treats multi-class registrations as independent observations, and ignores that gate outcomes correlate within owner. The primary estimates therefore come from a linear probability model on 3.10 million unique registrations (one row per serial), with ΔKL quintiles cut *within* class×cohort cells, class×cohort fixed effects, controls for counsel of record, intent-to-use basis, log description length, log owner filing volume, owner domicile, and foreign filing basis, and standard errors clustered on 1.27 million normalized owners (Table 2).

Three properties of Table 2 matter. The gradient is monotone within cells (+0.44, +0.90, +1.31,

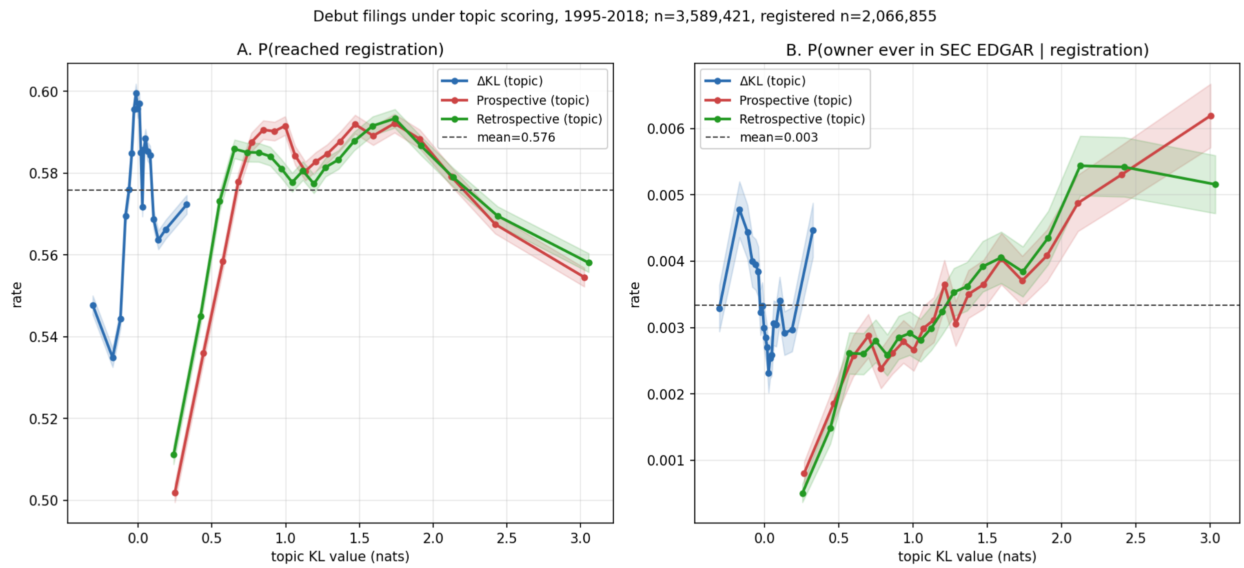

Figure 2: Outcomes for debut filings (owner's first-ever filing) across all 45 Nice classes under topic scoring, filing years 1995–2018, in quantile bins with 95% intervals. $n = 3{,}589{,}421$; registered subset $n = 2{,}066{,}855$. *A:* P(reached registration). *B:* P(owner ever in SEC EDGAR | registration); the structure in panel B is decomposed in Section 4.3.

+2.29pp across quintiles two through five) and essentially unmoved by the full control set, so it is not absorbed by who files or how they draft. It is, if anything, *largest* in the cleanest stratum — counsel-represented, US-domiciled owners — so it is not carried by the self-filed or foreign mass-filing segments whose gate failure rates are high for their own reasons.[3] And about forty percent of the naive pooled contrast was composition: the design-based magnitude is +2.2–2.6pp, not the raw +3.9pp.

The penalty is modern. It is indistinguishable from zero for the 2002–2007 cohorts, switches on over 2008–2010, and runs +4 to +9pp for every cohort since (Figure 4).

That timing invites an out-of-sample check, and the 2019 cohort supplies one: it is the first whose gate window falls entirely after the pandemic, and it reproduces the recent-cohort magnitude at +9.1pp raw among fully-elapsed registrations ($n = 313{,}725$). The 2020–2021 gates have not yet elapsed at the data edge.

The filer population also changed over exactly this period, which is why the cohort thirds of Table 2 matter. *Within* counsel-represented US-domiciled owners, and with the full control set, the gradient still rises from +1.0pp (2002–07) to +3.7pp (2008–12) and +3.1pp (2013–18). The forward-leaning gate risk is therefore a feature of the post-2008 filing economy itself, not of the post-2015 surge in self-filed and foreign applications, and it predates the pandemic by a decade.

Levels and gradients separate cleanly across filer strata. Self-filers fail the gate far more often than represented owners (56.8% versus 43.0%), and use-based registrations more often than intent-to-use ones, but the ΔKL gradient itself is similar across these strata once class, cohort, and drafting covariates are held fixed (+2.5pp self versus +2.1pp counseled in Table 2): who files predicts the level of gate risk; what the filing says predicts its increment. Per class, the largest penalties sit in

[3] The level effects on those covariates are large in their own right: counsel −16pp and Madrid-protocol basis −46pp. The latter is mechanical rather than behavioural, because §66(a) registrations file §71 rather than §8 declarations; they are flagged and controlled throughout.

Table 2: First-gate failure and ΔKL: design-based estimates. LPM of event-dated first-gate failure on within-class×cohort ΔKL quintiles (Q1 omitted), unique serials, registrations 2002–2018. All specifications include class×cohort fixed effects; controls are counsel of record, intent-to-use basis, log goods/services length, log owner filing count, owner domicile (US, CN), and foreign basis (§44(e), §66(a)). SEs clustered on normalized owner.

Sample	Specification	Q5−Q1 (pp)	t	n
All	FE only	+2.23	9.6	3,104,951
All	FE + controls	+2.29	7.8	3,104,951
Counsel-represented	FE + controls	+2.13	6.1	2,414,653
Counsel + US-domiciled	FE + controls	+2.63	6.3	1,888,655
Self-filed	FE + controls	+2.53	10.4	690,298
Counsel + US, 2002–07	FE + controls	+1.04	3.1	552,286
Counsel + US, 2008–12	FE + controls	+3.73	5.0	551,930
Counsel + US, 2013–18	FE + controls	+3.06	7.0	784,439
All	continuous, pp per σ	+0.87	9.8	3,104,951
Counsel + US	continuous, pp per σ	+0.94	7.3	1,888,655

tobacco (+26pp, the vaping era), scientific/tech services (+14pp), and lighting (+12pp, the LED transition), with small negative lifts in staple goods (jewelry, foods, alcohol).

The risk attenuates but persists at the year-ten renewal. In the software diagnostic class, where the event-dated first-gate lift on the 2010–2013 cohorts reaches +20.8pp, the conditional lift at the second gate falls to +7.0pp; pooled across classes the second-gate pattern is a shallow U (40.4% at the bottom quintile, 37.2% in the middle, 42.2% at the top), so by year ten the backward-leaning tail has caught up in mortality. Most of the ambition bet resolves at first proof of use. Prosecution telemetry sharpens the mechanism: among represented software filings, response speed to office actions predicts the gate (fastest-quintile responders pass at 57.4% against 49.4% for the slowest), and the ΔKL penalty survives within both fast and slow halves (−7.2 and −8.0pp), so the risk attaches to the product bet itself rather than to dying firms detected late.

Mixing the past and future reference windows identifies which kind of novelty the gate punishes. The penalty loads on breaking with the established past (−7.3pp) rather than on resembling the future (−3.2pp), and in software the separation is complete (−12.7pp against −0.3pp). Appendix A reports the full decomposition, plus a look-ahead-free months-scale variant that reproduces the pattern and a snapshot cross-check that agrees with the event-dated estimates under every scoring tried.

4.3 Surprising language marks firms that reach public markets; having been early does not

Panel B of Figure 2 contains two patterns, and they pull in different directions. Listing probability — P(owner ever in the SEC EDGAR universe), conditional on registration — rises monotonically in the KL *levels* on both axes (from 0.20% at the bottom prospective quintile to 0.51% at the top in raw bins, replicated at $T = 200$ and $T = 500$). And in signed ΔKL it is U-shaped: both tails beat the middle quintiles, with the lagging tail highest (0.41% at Q1, 0.27% at Q3, 0.34% at Q5).

The design-based estimates (Table 3) resolve both patterns into one decomposition. Atypicality pays, modestly: within class and filing year, holding description length fixed, top-quintile atypicality raises listing probability by +0.043pp on a 0.189% base — about a quarter more likely ($t = 3.8$).

Table 3: Debut listing and vocabulary position: design-based estimates. LPM of P(owner ever in SEC EDGAR) on 1,556,987 registered debut filings 1995–2018, one observation per owner, class×filing-year fixed effects, heteroskedasticity-robust SEs. Quintiles cut within class×year. Base rate 0.189%.

	Coefficient (pp)	t
Atypicality (prospective-KL level quintiles, Q1 omitted)		
Q5, FE only	+0.056	5.0
Q5, FE + log length	+0.043	3.8
Signed ΔKL quintiles, FE + log length + KL levels		
Q3	−0.038	−2.8
Q4	−0.064	−4.3
Q5	−0.068	−4.1
Decomposition, FE + log length		
$\lvert\Delta\text{KL}\rvert$ (per σ)	+0.024	3.8
signed ΔKL (per σ)	−0.019	−5.0
log description length	+0.044	14.6

Most of the raw 2.5-fold gradient is composition (listing-bound firms cluster in particular classes, years, and multi-class filings), which is why the honest magnitude is a quarter rather than a multiple. Forward lean costs, at equal atypicality: the signed component enters negatively (−0.019pp per σ, $t = -5.0$) while the unsigned component enters positively (+0.024pp per σ, $t = 3.8$). The U-shape in raw bins is exactly this pair: laggards collect the atypicality premium without the forward penalty; leaders collect it minus the penalty; the middle quintiles collects neither. Once the levels are controlled, the apparent recovery of the far-forward tail disappears entirely (quintile profile monotonically declining, Table 3 middle panel).

The two axes now tell one coherent story across both outcomes. Atypicality — writing about a product in language unusual for its category — is the differentiation signal: it marks real, specific, differentiated offerings and the firms that list. Forward lean — writing in language the category subsequently adopted — is the timing bet: it costs at the use-proof gate (Section 4.2) and it costs in listing at equal atypicality. Being early is expensive even when being different is valuable. The lifecycle and firm-level evidence agree on the sign of the bet; they simply price it at different points in the mark's life.

4.4 The construct is not patenting

If ΔKL and patent counts were noisy measures of the same underlying construct, they should correlate within firms over time. They do not. The panel used for this comparison scores ΔKL under an exponential-decay reference kernel rather than the flat window used elsewhere; the two correlate at $r = 0.87$ on 68,013 matched owner-years, so the contrast is not an artifact of the kernel. On a panel of 174,569 firm-years of trademark owners name-matched to PatentsView patent assignees (PatentsView, 2024) (94,106 firm-years across 14,470 firms with within-firm variation), a two-way fixed-effects regression of standardized firm-year mean ΔKL on log(1 + patents) gives -0.048σ ($t = -7.9$), strengthening under filing-count control (Table 4). Re-scored under the paper's method the distinctness sharpens: -0.014σ ($t = -2.3$) under topic scoring and -0.010 ($t = -1.5$) at $T = 200$ — essentially uncorrelated within firm, which is the property the construct needs. The strongest negative coefficient sits on patents$_{t-1}$: in years after a firm patents heavily, its trademark vocabulary regresses toward class-typical, consistent with temporal substitution between patent prosecution and market-facing brand work.

Two objections to reading that null as orthogonality are worth taking seriously, and both fail. The first is timing. Under a linear model of the innovation process a patent is applied for before its invention is in public use, while a trademark is filed when the firm is ready to sell, so patents would lead. Firms do not proceed that linearly, however, and in some industries the trademark application comes first (Seip et al., 2018), so the ordering is an empirical question rather than an assumption to build on. Entering patents at $t-2$, $t-1$, t, and $t+1$ — the last allowing the trademark to precede the patent — leaves nothing at any timing: −0.015 at two years out ($t=-1.61$), −0.009 at the lead year ($t=-1.08$), −0.010 contemporaneous ($t=-1.51$), and +0.003 with the trademark first. No ordering recovers a co-movement.

The second objection is heterogeneity: a pooled near-zero is consistent with genuine orthogonality everywhere, but equally with a positive slope in sectors where the two rights are complements — pharmaceuticals, chemicals, medical devices, machinery — offsetting a zero or negative slope where trademarks do the work alone. Splitting the panel on a complement/neutral/substitute grouping fixed in advance, and re-estimating the identical specification inside each cell, gives −0.000 ($t=-0.03$) among complements, −0.026 ($t=-1.45$) among substitutes, and an interaction of −0.008 ($t=-0.49$). Ordering all 27 Nice classes that clear a 50-owner floor by coefficient, the complement classes run from sixth place to twenty-fourth and do not agree in sign with one another: pharmaceuticals +0.011, software and electronics +0.003, medical devices −0.002, chemicals −0.070, machinery −0.090. The classes that do reach significance run against the complementarity story rather than merely failing to support it — the only significant positives are transport and cosmetics, neither a complement class, while the significant negatives include medical devices and machinery, both of which are. One caveat cuts in the construct's favour: because the panel is conditional on a successful assignee match, every owner in it patents at least once, so the substitute cells hold patenting firms in low-patenting sectors rather than typical firms in those sectors. That makes the groups more alike than they are in the population and biases the interaction toward zero, so the cell-level nulls, not the interaction test, carry the weight. This is the sample in which a patent–vocabulary relationship should be easiest to see, and it is not there in any cell.

Table 4: Within-firm ΔKL versus patenting. Coefficients in standard deviations of ΔKL per unit $\log(1+\text{patents})$.

Specification	coef (σ)	SE	t
Between-firm correlation (firm means)	−0.020	—	—
Within-firm correlation (firm+year demeaned)	−0.015	—	—
Two-way FE, contemporaneous	−0.048	0.006	−7.9
robustness: patents = max over matched assignees	−0.048	0.006	−8.0
controlling log(n_filings)	−0.051	0.006	−8.4
Lead/lag: patents_{t-1}	−0.054	0.009	−5.8
patents_{t}	−0.019	0.010	−1.9
patents_{t+1}	+0.010	0.010	+1.0

Firms on BCG/MIT "most innovative" lists sit above the panel mean under every scoring, but on 41 matched firms the differences are too imprecise to carry weight (Appendix A).

4.5 Each axis bites at a different stage

The outcomes above are estimated in separate places on separate samples, which obscures the pattern that matters most: the two axes do their work at different points in a mark's life, and where each one bites is informative. Table 5 re-estimates every outcome on one harmonized sample — one row per debut owner, scored on that owner's first filing — under a single specification, so the coefficients can be read down a column. Because it harmonizes the sample, its magnitudes differ from the section-specific tables above, which each use their own; the signs and the ordering are what the table is for.

Table 5: Every outcome in the funnel, conditioned on its prerequisite, under one specification. Unit is the debut owner, scored on that owner's first filing. Each row is a linear probability model with class×debut-year fixed effects and heteroskedasticity-robust errors; coefficients are percentage points per standard deviation. *Atypicality* is the KL level against the past; *forward lean* is the signed difference. Cohort windows differ because the Section 8 gate needs elapsed time and Form D is electronic only from 2009, so the rows are not a nested decomposition.

				Unsigned		Signed
Outcome	Given	n	Base	KL level	$\lvert\Delta\mathrm{KL}\rvert$	Forward lean
Reached registration	filed	3,161,403	56.12%	−0.906 (-28.3)	−0.344 (-7.4)	+1.186 (+41.3)
Passed first S8 gate	registered	1,774,111	50.94%	+1.216 (+28.5)	−0.631 (-10.2)	−1.351 (-35.5)
Raised a Reg D round	filed	1,534,824	2.53%	−0.023 (-1.6)	+0.034 (+1.1)	−0.270 (-15.4)
Raised a Reg D round	registered	916,442	2.61%	−0.056 (-2.8)	−0.015 (-0.4)	−0.229 (-10.1)
Listed after the round	funded	38,886	3.49%	+0.447 (+3.6)	+0.618 (+3.0)	−0.686 (-5.4)
Reached listed universe	registered	1,774,111	0.23%	+0.030 (+6.6)	+0.008 (+1.1)	−0.005 (-1.1)

t-statistics in parentheses.

Two reversals stand out, and both turn on the difference between the examiner and the market. Surprising language costs at registration and pays afterwards: atypical debut filings complete registration less often (−0.91pp per standard deviation), then survive the first use-proof gate more often (+1.22pp) and reach public markets more often (+0.03pp on a 0.23% base). The examination process penalizes surprise; the market does not. Having been early runs the other way: forward-leaning applications complete registration *more* often (+1.19pp) and then fail the use-proof gate (−1.35pp). Those filers follow through on the paperwork; what does not follow through is the product.

The registration row needs one caution. A linear coefficient cannot represent the inverse-U documented in Section 4.2, where both vocabulary tails complete less often than the middle. The positive linear term reflects the asymmetry of that shape — the forward tail completes more often than the backward tail — and not a monotone gain from leaning forward.

The financing rows are the ones worth dwelling on, because atypicality changes sign between them. Owners whose debut filing is atypical are *less* likely to raise a Regulation D round: −0.06pp per standard deviation among owners who completed registration ($t = -2.8$), against a 2.61% base. Among owners who did raise one, atypicality then predicts *reaching* the public market: +0.45pp on a 3.49% base, about 13% relative ($t = 3.6$). Investors do not appear to price atypical language at entry, yet it separates the firms they funded that go on to list from those that do not. Selection running one way and realized outcomes the other is what a screen leaving something on the table looks like, and the text carrying the signal is free and public at the moment of filing.

That last row requires one construction that is easy to get wrong. An exchange-registration marker on its own does not mean a firm listed *after* raising: 56% of matched owners carrying one registered before their first Regulation D notice, because established public companies also place securities privately. Read naively, the row would be measuring public companies doing private placements rather than funded firms going public. It is therefore restricted to owners whose exchange registration postdates their first notice, which cuts the positive coefficient roughly in half from its uncorrected value and leaves it significant.

Three further limits bound how hard the reading can be pushed. Regulation D notices are electronically filed only from 2009, so the financing rows rest on 2009–2018 debut cohorts rather than the full panel. Owners are matched to filers by normalized name, which succeeds for a small and unrepresentative minority and favours formally constituted entities, so every financing estimate is conditional on matchability. And the final row conditions on funding, which is realized after the filing being scored; the coefficient is a contrast among survivors, not a treatment effect. What it establishes is that information distinguishing eventual listers from other funded firms was already present in the goods/services text years earlier — not that atypical language caused the outcome.

Forward lean, by contrast, does not reverse anywhere after registration. It is negative at the use-proof gate, negative at financing whether or not registration is imposed, and negative at listing among the funded. Being early is penalized at every stage where the market, rather than the examiner, is doing the selecting.

5 Business vocabulary flows out of software, and incumbents carry it

The spatial version of this question is answered: von Graevenitz et al. (2022) show that novel description tokens reach distant US locations more slowly than near ones, in the *Regional Studies* programme surveyed by Castaldi & Mendonça (2022). The cross-industry version is open, and the Nice class system supplies the partition.

Cross-industry idea diffusion is directly measurable on this corpus. For 14 curated business-model phrases, Table 6 records the first year each of four classes (software 009, tech services 042, transport 039, advertising/retail 035; 2.32 million granted filings) crosses a 1% prevalence floor. Software- and tech-services-origin phrases arrive in the physical-economy classes with lags of one to thirteen years; the only counterexample in the set is on-demand, which originates in transport. The 1% floor is sensitive to class size, and the asymmetry is a four-class result a 45-class build would test properly.

Table 6: First year a theme reaches 1% prevalence among granted filings, by class. “–” = never clears the floor.

Theme	software	tech-svcs	transport	advert/retail
as-a-service	2011	**2009**	2013	2012
mobile-app	2010	**2009**	2012	2012
cloud	2012	**2010**	2011	2014
streaming	**2008**	2008	2013	2014
artificial-intelligence	2017	**2016**	2018	2018
augmented/virtual-reality	**2015**	2016	2022	2022
blockchain / crypto	2021	**2018**	–	2022
real-time	**2001**	2003	2009	2014
on-demand	2017	2016	**2014**	2017

Unsupervised theme extraction corroborates the pattern without depending on which phrases were picked. A detection-purpose LDA ($T = 50$ on a stratified 200,000-filing sample) yields recognizable business-model components; for each (theme, class) cell crossing the 1% floor with six or more tracked years, Bass-model fits (Bass, 1969) on 118 retained cells give median innovation coefficient $p = 0.018$, median imitation coefficient $q = 0.110$, and median $q/p \approx 6.5$, in the range typically reported for consumer-product adoption. Tagging each multi-class theme's origin class and counting directed origin-to-arrival edges: software contributes 33 of 72 edges against 18 expected under symmetry, and receives only 10. The asymmetry reproduces under independently-fitted NMF on the same sample; both are bag-of-words factorizations, so the robustness is within-paradigm.

Incumbents, not entrants, carry the canonically new themes. The 50 earliest carriers per theme are 73.7% entrants on average — but the corpus's year-matched debut baseline is 76.4%, so the theme-carrier excess is −2.7pp and 28 of 50 themes sit *below* baseline. The themes leaning incumbent relative to baseline include environmental/sustainability, natural gas and petroleum, artificial intelligence, and cloud computing: established firms diversifying in pull the vocabulary signal forward at least as much as de-novo entrants do (Schumpeter, 1942). This base-rate discipline is general — population-share questions on this corpus mislead unless the debut rate is netted out.

The AI era, monitored this way, has so far compounded rather than erupted. Aligning era starts (internet 1994, AI 2015) and tracking era-term prevalence in the software and tech-services classes: through year three the two track closely (5.1% versus 4.0% of filings), then diverge — internet vocabulary tripled between years four and six, reaching 23% of filings in 2000, while AI vocabulary has compounded steadily to 12.9% at year nine, roughly where the internet stood at year five (Figure 5). The release of consumer chatbots is visible as an acceleration (6.8% in 2022 to 10.2% in 2023), not a discontinuity, and per-year corpus turbulence shows no AI-era deviation through 2024. Combined with the incumbent lean of the AI theme, the trademark record through 2024 reads as incumbent-led vocabulary adoption rather than an entrepreneurial explosion. The comparison is sensitive to the era-start alignment chosen, and it carries a falsifiable marker: if AI follows the internet's path with a lag, its prevalence should triple within roughly three years of the 2024 data edge, and the entrant share of its carriers should rise above the corpus baseline rather than sit below it.

6 What ΔKL does not measure

Four claims are defended. ΔKL, adapted from Barron et al. (2018); Murdock et al. (2017) onto topic distributions of USPTO trademark filings, produces interpretable structure on commercially-incentivised legal text. Vocabulary position carries two separable signals: atypicality marks listing-bound firms (+23% relative, within class-year with length fixed), and forward lean marks product risk (+2.3pp per quintile at the first use-proof gate with fixed effects, controls, and owner clustering; −0.019pp per σ in listing at equal atypicality). The construct is empirically distinct from patent-track invention within firm. And the corpus supports direct measurement of cross-industry vocabulary diffusion, with multi-year origin-to-arrival lags, Bass-fittable adoption, and strongly asymmetric flow out of software.

Three readings are rejected. ΔKL does not measure innovation in any strong sense; the defended primitive is lead/lag vocabulary position on commercial text, and Appendix D shows concretely which kinds of novelty the primary scoring cannot see. ΔKL does not identify an exploitable extreme-tail return premium; the large debut-specification figure fails a sample-size diagnostic. And the term-scored signed firm-level advantages of forward-leaning filings (margins, listing) are artifacts of drafting style and lexical specificity (Appendix B).

The main open questions are dated and specific. Why the gate penalty emerged with the

2008–2010 cohorts is unresolved: the within-stratum estimates rule out the post-2015 filing-surge composition story, but cohort and gate year move together, so post-crisis bet composition, app-era speculative filings, and calendar-time enforcement changes cannot yet be separated. A prospective-only variant of the construct, with reference vocabularies and topic bases refit on vintage data, would close the remaining look-ahead channel (the current representation is fit on the pooled corpus; the months-scale freshness variant is already look-ahead-free and reproduces the risk pattern in its diagnostic class). Beyond these: linking gate failure to observable commercial death (owner filing cessation, EDGAR exit); the 45-class diffusion build; theme-value portability across industries; the supplier-versus-deployer split within arriving themes; demographic enclaves and theme origination; validation against the USPTO ID Manual's codification history; and a cross-paradigm (embedding-based) theme extraction.

7 Reproducibility

Code, firm-year panels, and analysis outputs are available at `https://github.com/ericsilver/novelty`. The pipeline rebuilds from a USPTO Open Data Portal API key with a single `make tm` command (the underlying TRTYRAP backfile is approximately 12 GB streamed once). The published panels include the SEC-matched firm-year ΔKL panel ($n = 59{,}033$) and the PatentsView-joined panel ($n = 174{,}569$). Analysis scripts and their outputs are indexed in the repository `README.md`.

A Computation, corpus, and robustness

A.1 Corpus and scoring detail

The USPTO Trademark Annual Applications backfile (product TRTYRAP) provides a complete record of US trademark filings from 1884 onward. The deepest diagnostic work uses the complete software/electronics class (Nice 009; 1.81M filings); the diffusion application uses four classes totalling 2.32 million granted filings 1990–2024; the gate analyses use all 45 classes. The firm-level panels link trademark owners to SEC EDGAR financial filers and to PatentsView patent assignees by normalized-name join.

The topic representation is a $T = 50$ LDA fit on a stratified sample of 448,437 filings across all 45 classes (vocabulary of 62,168 unigrams and bigrams); every filing 1990–2024 is transformed to a dense 50-topic distribution, and prospective and retrospective KL are computed against class-year topic references at $W = 5$, scoring 1995–2019. A term-level variant is retained alongside: token distributions (unigram + bigram, within-class document frequency ≥ 50) against the same class-year windows with Dirichlet smoothing ($\alpha = 1$), computed at $W \in \{3, 5, 7\}$. The term variant is used where vocabulary identity is the object (phrase transit, the window decomposition, the freshness measure) and as the comparison scoring throughout.

A.2 Burn-in and interpretable range

The signal depends on having enough past and future filings to populate the references; the 5-year past window is incomplete before 1990 and the future window after 2020, and both zones are excluded (Figure 6). The burn-in cut is set empirically: thin past references inflate prospective surprise, and profiling that bias across all 44 sufficient-volume classes gives a median absolute bias of $\sim$2.0 nats in 1990, 0.25 in 1991, 0.10 in 1992, and 0.04 from 1993 onward, below any interpreted effect size. The 1995–1997 deviations ($\sim$0.15–0.17) are not burn-in — mechanical contamination cannot rebound — but the dot-com era moving the corpus. Per-class burn-in lengths estimated under a stability-band criterion track each class's noise floor rather than its reference density, so a single standard cut at 1995 is used, conservative by about two years.

A.3 Reliability and fragilities

Prospective and retrospective KL are highly correlated however scored (term: $r = +0.971$ on the complete class-009 corpus; topic: 0.950), so signed ΔKL is a small residual of two large, stable quantities. Under term scoring the residual inflates mechanically for short filings (mean $|\Delta\text{KL}|$ 2.5-fold across length bands); under topic scoring the gradient falls to $\sim$1.4-fold. The two scorings agree only moderately per filing (Pearson 0.46, Spearman 0.36 on 1.9M jointly scored observations), which gives their agreement on the mark-level patterns evidential force and their disagreement on the firm-level patterns diagnostic force (Appendix B). Across $W \in \{3, 5, 7\}$, 6.9–10.4% of filings flip sign. Per-filing noise attenuates at every aggregation used: firm-year means, token pools, theme prevalence trajectories.

Alignment with expert innovation judgment is at most suggestive. Firms on BCG/MIT "most innovative" lists sit $+0.09\sigma$ above the panel mean under term scoring ($p = 0.22$), $+0.14\sigma$ under topic scoring ($p = 0.07$), and $+0.08\sigma$ at $T = 200$ ($p = 0.20$). The matched sample is 41 firms, which is too small to carry weight in either direction, and the paper draws no inference from it. Patent counts remain uncorrelated with ΔKL on the same panel (Pearson +0.010).

A.4 Scoring robustness of the mark-level results

The snapshot cross-check on the single one-gate-elapsed cohort (registrations 2016–2018, terminal status 710 versus 701–705/800) agrees with the event-dated estimates and is scoring-robust: -5.5pp survival across quintiles under topic scoring, -5.7pp under term scoring on the identical sample, -5.5pp at $T = 200$ (Figure 7). Under term scoring, gate survival *rises* in the raw KL level (+6.5pp prospective, +7.6pp retrospective): atypical text survives the use requirement better while forward-leaning text survives it worse — the same two-axis separation as Section 4.3, at the mark level.

A.5 Which novelty: the reference-window decomposition

"Forward-leaning" is relative to a reference, and mixing windows across the two sides identifies which novelty the gate punishes. Scoring the corpus at $W \in \{3, 5, 7\}$ symmetric plus a foresight configuration (prospective against the recent 3-year past, retrospective against the distant 7-year future) and a past-rupture configuration (the reverse): the top-versus-bottom gate lift runs -3.2pp for foresight, -4.7pp at symmetric $W = 3$, -5.7pp at the $W = 5$ baseline, -6.1pp at $W = 7$, and -7.3pp for past-rupture, the most pervasive variant (negative in 35 of 41 classes; Figure 8). In software the decomposition is complete: foresight carries no penalty (-0.3pp) while past-rupture carries -12.7pp. Section 8 risk attaches to breaking with the established past, not to resembling the future, consistent with vocabulary leaving the corpus gradually: the long-past reference holds the fading establishment, and distance from it is what the use requirement punishes. The registration inverse-U is indifferent to the window mix (+4.6 to +4.9pp depth across all five variants), so the examiner and the marketplace respond to different aspects of the same text.

The decomposition extends below the year. Scoring class 009 monthly against two adjacent references (the twelve months ending one month before filing, and the twelve before those), the two levels are nearly identical per filing ($r = 0.992$) but their difference — a months-scale freshness measure, high when a filing resembles the very recent months far more than the year before them — reproduces the risk pattern at -9.4pp across quintiles, stronger than the symmetric baseline in the same class, with the penalty concentrated on the months-late rather than the months-early. Freshness is built entirely from pre-filing text and is computable on the filing date with no look-ahead; corpus-wide estimation is owed.

B The measurement hazard: term scoring manufactures firm-level signal

Under term-level scoring, forward-leaning filings appear to belong to better firms. Among registered debut filings, the probability that the owner ever appears in SEC EDGAR rises in term-ΔKL from 0.29% at the bottom quintile to 0.42% at the top; on an SEC-matched firm-year panel, trailing 3-year mean term-ΔKL predicts gross margin at +2.3pp per σ ($t = 2.7$ on the locally rebuilt panel through 2019; +3.2pp, $t = 5.3$, on the published panel through 2024), with prospective surprise entering positively and retrospective negatively; and firm-year term-ΔKL predicts excess stock returns of +1.5 to +5.0pp per σ at one-to-four-year horizons. A larger debut-only return figure (+28pp at 4 years) fails a sample-size diagnostic and is not claimed.

None of the signed versions survives distribution scoring. On identical samples, the debut listing gradient in signed topic-ΔKL is flat to declining, and the margin coefficient reverses: -1.3pp per σ ($t = -1.8$) at $T = 50$, -2.2pp ($t = -3.1$) at $T = 200$, and -0.8pp ($t = -1.1$) at $T = 500$, against the term-level +2.3pp; the prospective/retrospective decomposition flips sign and stays flipped. The term-level listing gradient separately dissolves under document-length controls: within length bands it is U-shaped rather than rising, with the monotone rise contributed by composition across lengths (listed firms' counsel write longer descriptions). The mark-level results are unaffected — they hold under both scorings at every resolution.

Vocabulary blindness does not rescue the signed signal. The topic model's vocabulary is sample-built, so rare and emergent terms are invisible to it; if the firm signal lived exactly there, topic scoring would erase a real signal rather than an artifact. Computing each filing's invisible-vocabulary share (63.8% of class-009 term *types* but only $\sim$9% of token *mass*): the share itself is a strong firm marker (P(EDGAR) 2.9% to 5.5% across its quintiles; gate survival +5.7pp), but *within* invisible-share strata the signed ΔKL gradient on the firm outcome is flat to negative, while the gate penalty holds in both strata (-6.9 and -8.3pp). The term-level firm signal is *unsigned specificity*: serious firms describe specific products with specific, therefore rare, words, and rare-term mass inflates term-ΔKL mechanically. Topic count is a partition dial, not a coverage dial — a crypto/blockchain topic exists at $T = 200$ but not at $T = 50$ or $T = 500$, because raising T re-partitions the fixed vocabulary along dominant axes — so resolution sweeps test granularity, not coverage, and the invisible-share split is the test that binds.

The registration filter does not explain the divergence. Dropping the registration conditioning leaves the term-level listing gradient essentially unchanged (0.26% to 0.38% across quintiles against 0.29% to 0.42% conditional), and an examiner channel touching 4.3% of failed applications cannot generate several-point quintile gaps.

The portable conclusion: term-resolved text-novelty measures can manufacture firm-performance correlations from drafting style and lexical specificity, and a distribution-based rescoring is a cheap, decisive diagnostic. Four completions are owed: rescoring topics on the full per-class term vocabulary; a future-blind vintage refit of the topic model; decomposing term-ΔKL into in-vocabulary and out-of-vocabulary contributions and re-estimating the firm panels on each; and an emergence-event design comparing carriers of to-be-emergent terms against class-, year-, and length-matched controls with attorney-of-record fixed effects.

C The outcome analysis without registration conditioning

Figure 9 repeats the lifecycle outcome analysis without the registration conditioning (term-scored, where the composite shape was originally encountered), on filings 2013–2015 pooled across all classes (a window whose registrations land by ∼2017 and face at least one Section 8 gate by the 2025 snapshot; the earliest registrations in the window also face their year-ten renewal, so panel B mixes one and two gates). Panel A is the unconditional composite, the probability that a filing both registers and passes the gate, measured over all filings including the never-registered. It is an inverse-U in ΔKL (quintiles 22.0%, 22.9%, 23.0%, 22.5%, 21.4%; middle-versus-tails depth +1.4pp, SE 0.1pp), because the composite inherits the registration step's middle-favoring shape from Figure 2A. Unconditional survival analyses on this corpus therefore recover an inverse-U whether or not the maintenance gate itself favors the middle, which is why the body of the paper conditions on registration throughout. Panel B shows the conditional gate outcome on the same filings for contrast; on this mixed one-to-two-gate cohort it is mildly inverse-U with a clearly penalized forward tail, consistent in the right tail with the cleaner single-gate cohort of Figure 7 and a reminder that the left tail of the gate result is cohort-sensitive.

D How three filings encode

Table 7 traces three filings through term scoring and topic scoring at $T \in \{50, 200, 500\}$, illustrating three distinct kinds of novelty and how each representation handles them.

Amazon.com (1997, advertising/retail) is *combinatorial* novelty: every word is ordinary, but the bigrams assemble an invention ("computerized line," "line search," "search ordering"; class document frequencies 379, 134, 355). The filing's text reads "on line" as two words, the standard spelling in 1997, before *online* lexicalized into a single token; the analyzer drops the stopword "on" and the surviving bigram "line search" is the act of forming the compound caught in progress. The novelty was necessarily combinatorial because the word for it did not yet exist. Term scoring sees these bigrams; under topic scoring at every resolution all of them are out of vocabulary and the filing reads as a media retailer that uses computers.

Kombucha (1997, beer/soft drinks) is *lexical* novelty: the new thing arrives as new words (*kombucha* df 567, *saccharose* out of vocabulary even at the class level). The word *kombucha* is in the topic vocabulary but, with no kombucha-like topic, its mass dissolves into a generic beverages topic (≥ 0.60 weight at every T).

Ethereum (2018, software) is *wave* novelty: emergent vocabulary already diffusing (*blockchains*, *distributed computing*). It loads on generic software and services topics at every resolution, including $T = 200$, which does contain a cryptocurrency/blockchain topic — the eleven occurrences of *software* outvote the blockchain terms.

The pattern across the row is the measurement point in miniature: topic scoring cannot represent combinatorial novelty (Amazon), dilutes lexical novelty (Kombucha), and majority-votes away wave novelty (Ethereum) even when a fitting topic exists. That the mark-level outcome results are nearly identical under both scorings despite this blindness is what licenses topic scoring as a robustness instrument; that the firm-level results diverge is what exposes them as artifacts of the lexical specificity term scoring additionally captures (Appendix B).

Table 7: Top topic loading for three filings, by representation. Term scoring resolves the novel content of all three; topic scoring resolves none of it at any tested resolution.

	AMAZON.COM (1997)	KOMBUCHA (1997)	ETHEREUM (2018)
novel content (term)	line search, search ordering	kombucha, saccharose	blockchains, distributed computing
$T = 50$	video, computer, music (0.40)	beverages, fruit, drinks (0.64)	software, online, downloadable (0.33)
$T = 200$	entertainment, video, games (0.29)	beverages, drinks, coffee (0.60)	field, services, development (0.37)
$T = 500$	computer, software, data (0.22)	beverages, drinks, fruit (0.61)	software, online, computer (0.49)

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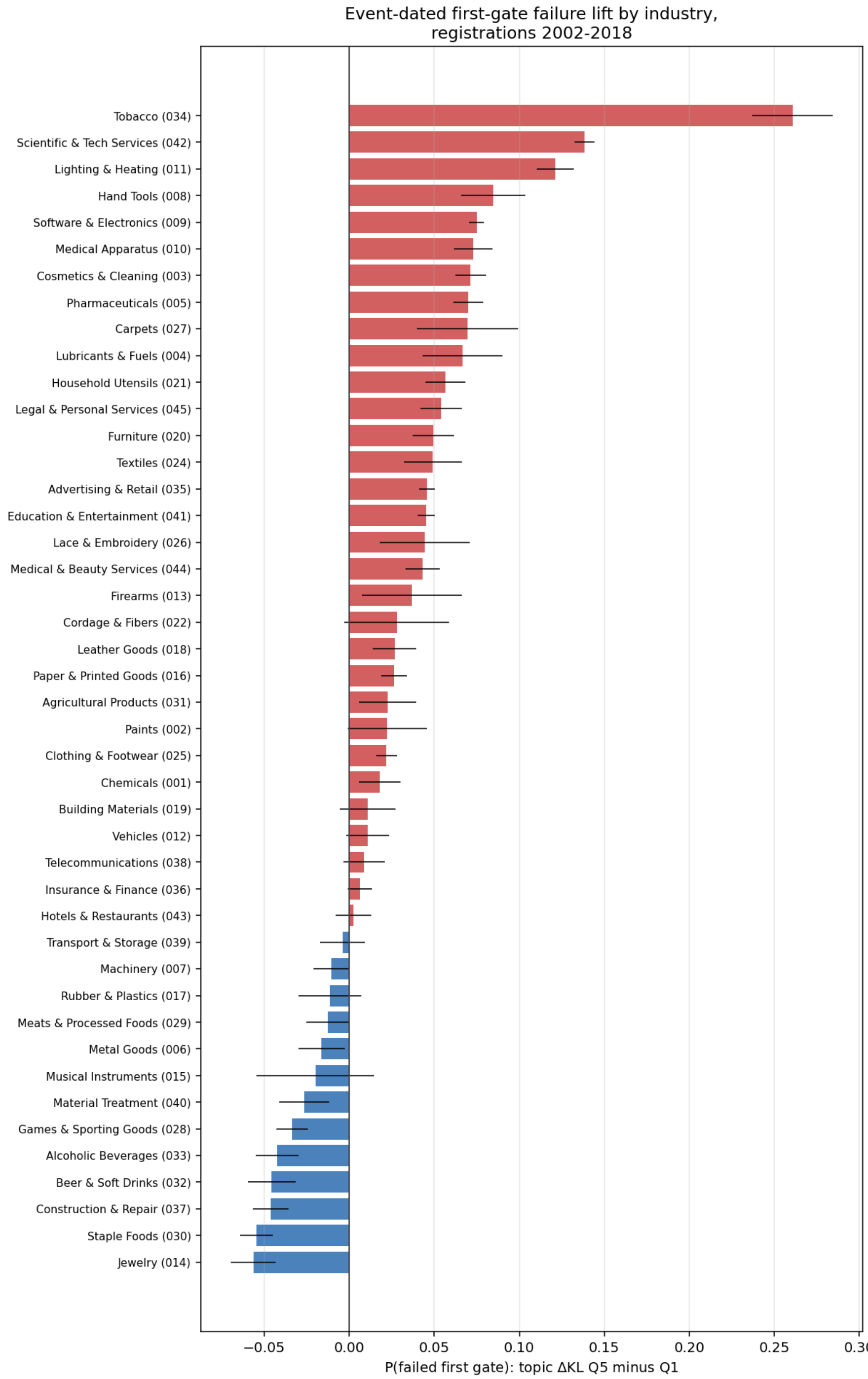

Figure 3: Event-dated first-gate failure lift (topic ΔKL top minus bottom quintile) by industry, registrations 2002–2018, raw quintile contrasts with 95% intervals. Positive bars indicate the forward-leaning penalty.

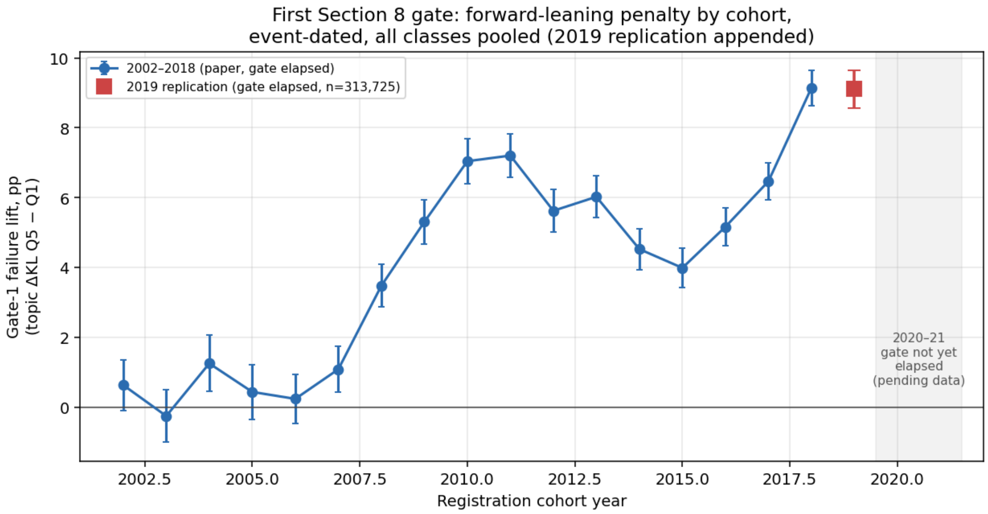

Figure 4: The forward-leaning gate penalty by registration cohort, event-dated, all classes pooled, raw quintile contrasts. The penalty emerges with the 2008–2010 cohorts and persists through every cohort since; the 2019 point is the out-of-sample replication on fully-elapsed registrations, and the 2020–2021 gates have not yet elapsed.

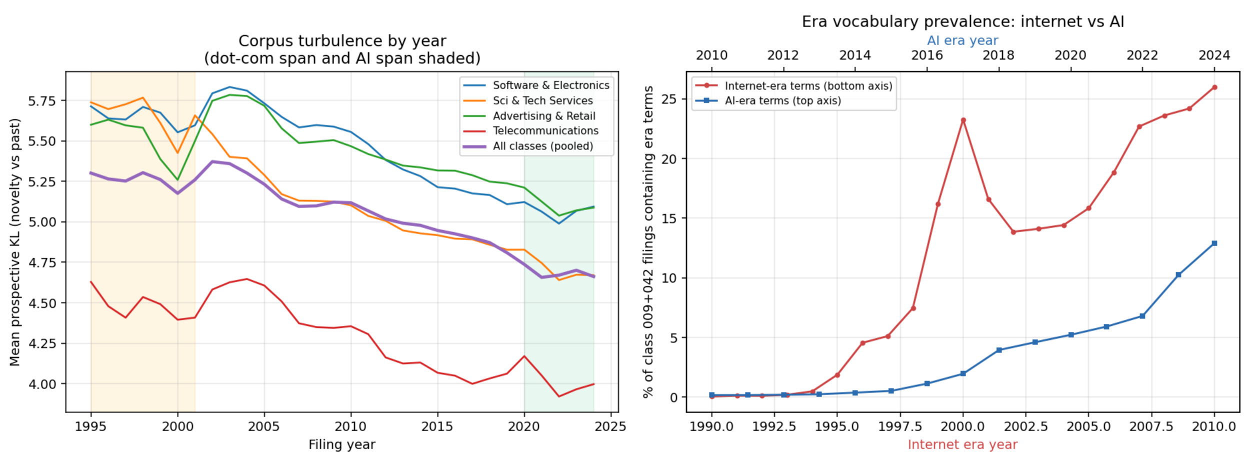

Figure 5: *Left:* per-year mean prospective KL by class and pooled, 1995–2024, with the dot-com and AI spans shaded. *Right:* era-term prevalence in classes 009+042, internet era (bottom axis, from 1994) against AI era (top axis, from 2015).

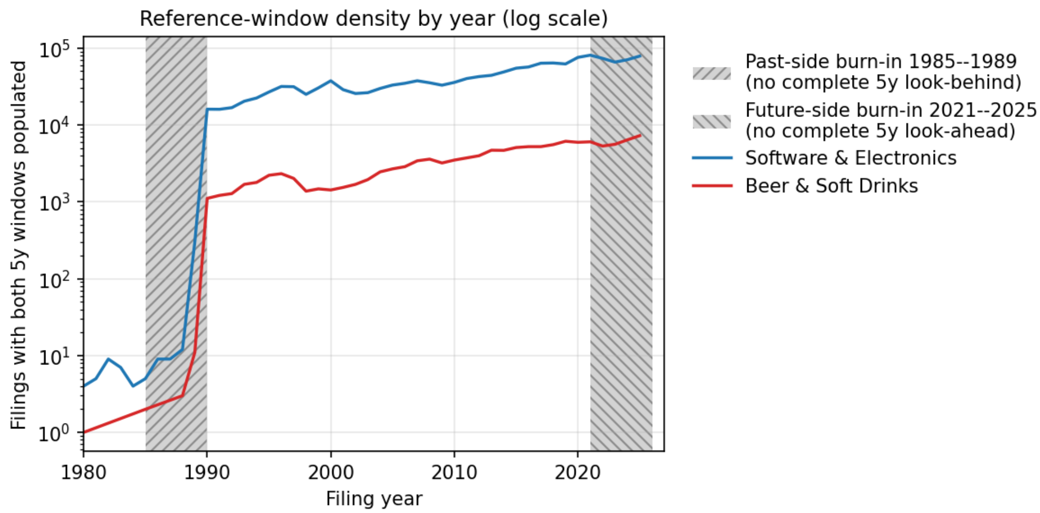

Figure 6: Reference-window density by year on log scale. Hatched regions are the past-side and future-side burn-in zones where the 5-year window is incomplete.

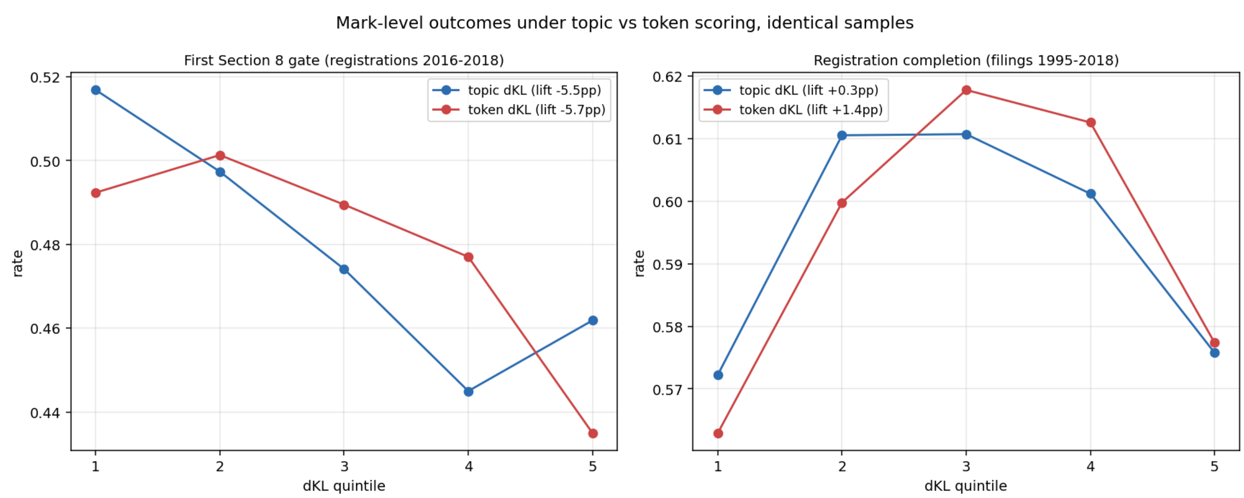

Figure 7: The two mark-level outcomes under topic and term scoring on identical samples, pooled across classes. *Left:* first Section 8 gate survival (snapshot, registrations 2016–2018). *Right:* registration completion (filings 1995–2018). Both patterns are scoring-robust.

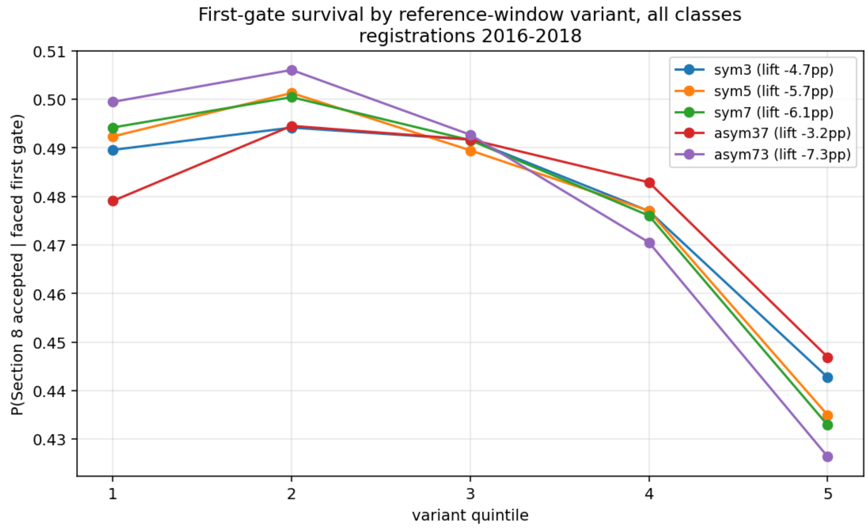

Figure 8: First-gate survival by quintile of each reference-window variant, pooled across classes, registrations 2016–2018. The penalty deepens monotonically from the foresight configuration (pros $W=3$, retr $W=7$) to the past-rupture configuration (pros $W=7$, retr $W=3$).

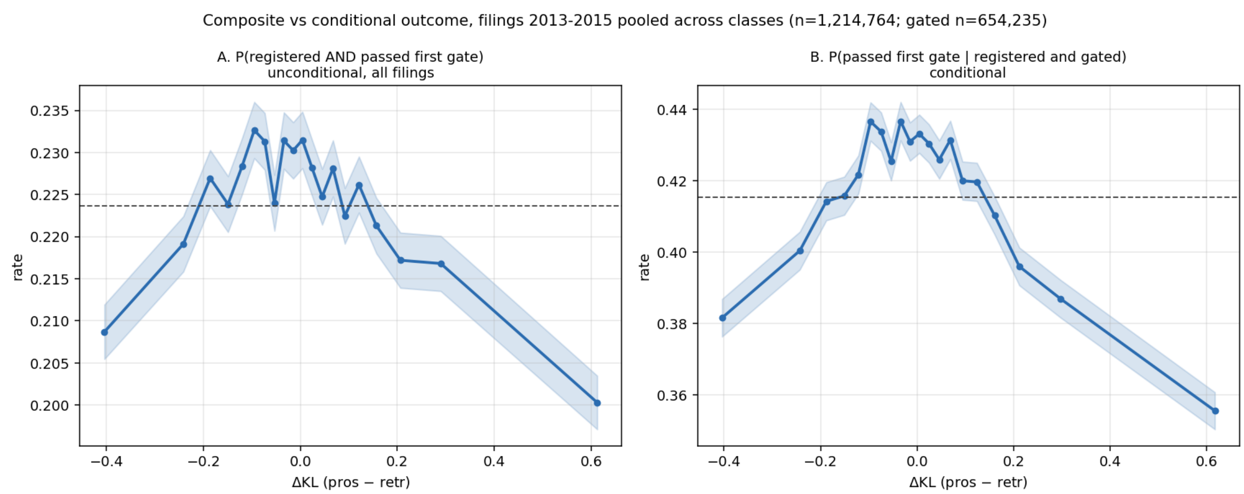

Figure 9: Filings 2013–2015 pooled across classes (n = 1,214,764; gated subset n = 654,235). *A:* the unconditional composite P(registered and passed first gate) over all filings. *B:* P(passed | registered and gated) on the same filings.